

# linear algebra Quiz 1

5문제, 30분

반번호이름 : \_\_\_\_\_

1. Prove the theorem. [10points]

For an  $n \times n$  matrix  $A$ , the following are equivalent:

- (a)  $A$  is invertible.
- (b)  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution.
- (c) The reduced row echelon form of  $A$  is  $I_n$ .
- (d)  $A$  is a product of elementary matrices.
- (e)  $A\mathbf{x} = \mathbf{b}$  is consistent for every  $n \times 1$  matrix  $\mathbf{b}$ .
- (f)  $A\mathbf{x} = \mathbf{b}$  has exactly one solution for every  $n \times 1$  matrix  $\mathbf{b}$ .

2. Prove the theorem. [10points]

A system of linear equations has zero, one, or infinitely many solutions.

3. Prove the theorem. [10points]

Every linear transformation from  $R^n$  to  $R^m$  is a matrix transformation, and conversely every matrix transformation is a linear transformation.

4. Write the adjoint matrix of  $A$  and prove the theorem. [10points]  
If  $A$  is an invertible  $n \times n$  matrix, then  $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$

5. Prove the theorem. [10points]

If  $A\mathbf{x} = \mathbf{b}$  is a system of  $n$  equations in  $n$  unknowns with  $\det(A) \neq 0$ , the unique solution is

$$x_1 = \frac{\det(A_1)}{\det(A)}, \quad x_2 = \frac{\det(A_2)}{\det(A)}, \quad \dots, \quad x_n = \frac{\det(A_n)}{\det(A)}$$

where  $A_j$  is the matrix obtained by replacing the  $j$ th column of  $A$  by  $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$ .